

100kW module, and achieves the 1MW, 1 Rack principle with a footprint that is more than halved.

This 100kW UPS power module will serve customers in the data centre industry. It combines ultra-high density FusionPower2.0 with Huawei SmartLi intelligent li-ion batteries to save more than 70 per cent footprint. SmartLi's unique active current balance technology supports the hybrid use of old and new battery strings. The SmartLi uses the safest LFP battery cell in the industry, which cannot catch fire due to thermal runaway. SmartLi's unique smart voltage balance technology ensures that the system can still run properly even if one battery module is faulty.

Huawei Technologies Co., China
www.huawei.com

Buck-boost regulator

Vicor has released its PI3740 ZVS buck-boost regulator with an extended operating temperature range of -55°C to +115°C and optional tin-Lead BGA



packaging. The PI3740 is a high-density and high-efficiency buck-boost regulator with an 8V to

60V input voltage range and supports output voltages from 10V to 50V. The device offers up to 140W in a 10mm × 14mm SiP package. Higher power can be obtained with additional devices in parallel. The ZVS switching topology also enables efficiency as high as 96 per cent.

Vicor Corporation
www.vicorpower.com

Battery management system

STMicroelectronics has launched battery-management controller L9963 that increases reliability, safety, driving range and cost-effectiveness of electric vehicles (EVs). It provides accurate voltage range for longer driving and battery life, along with temperature-monitoring inputs for enhanced safety. The device can monitor up to 14 stacked battery cells, measuring voltages with accuracy better than 2mV in the 1.7V-4.7V range to maintain superior cell condition. Simultaneous digitisation of all acquired

values eliminates cell-synchronisation delays.

The L9963 can also monitor up to seven external temperature sensors. Its BMS application uses several L9963 devices to monitor cell groups in the battery stack, managed by a host microcontroller such as ST's SPC5 high-performance, safety-critical automotive-qualified MCUs. With the help of a serial peripheral interface (SPI), the L9963 features a 2.66Mbps vertical communication interface for high-speed communication between multiple L9963 ICs for monitoring the entire battery stack. Thus, eight devices can convert and read 96 cells in less than 4ms.

STMicroelectronics Private Limited, Noida, UP
www.st.com

TEST & MEASUREMENT

Spectrum analysers for 5G

Siglent Technologies presents two spectrum analysers to meet the growing demand of high bandwidths for 5G applications.



SSA3075X Plus

SSA3075X Plus has a frequency range of up to 7.5GHz for delivering reliable automatic measurements and has multiple modes of operation. Optional functions are RF power measurement, vector signal modulation analysis, reflection measurement and EMI test. Applications include broadcast monitoring/evaluation, site surveying, S-parameter measurement, analogue/digital modulation analysis, EMI pre-compliance test, research and development, education, production, and maintenance.

SVA1075X also has a frequency range of up to 7.5GHz, comes with multiple modes of operation and delivers reliable automatic measurements. Its optional functions are distance-to-fault locator, a vector signal modulation analyser.

Siglent Technologies
https://siglenta.com

IoT modules testing

Rohde & Schwarz, in association with Gemalto, will be reducing expensive and

time-consuming drive tests. This collaboration enables both companies to help each other in the verification of software builds and releases. So, manufacturers of IoT solutions will be able to use virtual drive tests during the development phase of CAT M1 and NB IoT modules to find and fix problems at an earlier stage. This will also enable seamless cellular coverage and reliable connectivity before the integration process continues and further field tests are performed.

The setup consists of R&S Field-to-Lab wireless communication test system, R&S CMWcards smart network emulator and R&S CMW500/CMW290 wideband radio communication tester. The LTE-based R&S Field-to-Lab solution supports LTE-M and NB-IoT technologies. With the same user interface, customers can seamlessly move between LTE and IoT testing.

Rohde & Schwarz, Delhi
www.rohde-schwarz.com

MISCELLANEOUS

Lighting for smart LEDs

By taking advantage of its Power over Ethernet (PoE) technology and ultra-low-power Bluetooth Low Energy connectivity, along with AC-DC and DC-DC power conversion design, ON Semiconductor has introduced a connected lighting platform for engineers to explore IoT-based connected lighting. It features multiple forms of connectivity, such as Bluetooth Low Energy, that is provided by the RSL10 SIP, and Power over Ethernet. It offers up to 90W of output power and two independent channels supply and controls up to 16 LEDs.

This high-efficiency platform also enables the development of battery-less LED lighting applications when used together with the Energy Harvesting Bluetooth Low Energy Switch. It can control LEDs wirelessly using the RSL10 Sense and Control app (available for iOS and Android), or through a web client. Using the app, firmware updates can be easily sent after the application has been deployed.

ON Semiconductor, Bengaluru
www.onsemi.com

